Insights into the Oxygen Evolution Reaction on Graphene-based Single-atom Catalysts from First-principles Informed Microkinetic Modeling

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Abstract

Single-atom transition metals embedded in nitrogen-doped graphene have emerged as promising electrocatalysts due to their high activity and low material cost. These materials have been shown to catalyze a variety of electrochemical reactions, but their active sites under reaction conditions remain poorly understood. Using first-principles density functional theory calculations, we develop a pH-dependent microkinetic model to evaluate the relative performance of transition metal catalysts embedded in fourfold N-substituted double carbon vacancies in graphene for the oxygen evolution reaction. We find that reaction pathways involving intermediates co-adsorbed on the metal site are preferred on all transition metals. These pathways lead to enhancements in catalytic activity and broaden the activity peak when compared with purely thermodynamics-based predictions. These findings demonstrate the importance of investigating reaction pathways on graphene-based catalysts and other 2D materials that involve metal active centers decorated by spectator intermediate species.

Keywords: density functional theory, graphene, microkinetic model, oxygen evolution reaction, single-atom catalyst

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Introduction

Single-atom transition metals embedded in nitrogen-doped graphene have been identified as a promising class of materials that combine low material cost with high electrical conductivity and chemical stability. To date, these graphene-based single atom catalysts have shown high activity towards a broad range of electrochemical reactions including the hydrogen evolution[1–8], carbon dioxide reduction[9–15], and oxygen reduction[16–25] and evolution reactions[3,26–34]. Despite recent advancements, detailed atomic-scale understanding of the factors contributing to the enhanced activity for these materials, including their complex electrochemical interface, is still lacking. Hence, fundamental mechanistic studies could shed light on the *in situ* nature of the active site[35–42] and help inform catalyst design strategies.

Density functional theory (DFT) has become invaluable in assessing active transition metal based single-atom catalysts embedded in graphene-based systems for electrochemical applications[43–47]. In these studies, thermodynamic analyses are often performed to estimate onset potentials based on the most endergonic reaction step. While this methodology is convenient due to its simplicity, onset potentials obtained through such analyses often deviate substantially from experimental observations, both in the absolute magnitude of predicted onset potentials and in general trends between different systems. One possible factor contributing to these discrepancies is an accurate description of the active site under reaction conditions. The complex liquid-solid interface combined with single-atom nature of these materials makes conventional characterization techniques insufficient. These deviations have also been attributed to shortcomings of computational methods[45,46], however, recent work has indicated that these discrepancies cannot be solely attributed to limitations of the computational methods[24,40,48–51]. Incorrect description of the active site and relying solely on thermodynamic analyses for predicting activity can explain such discrepancies. For example, it was recently suggested that the presence of co-adsorbed oxygen species present during the oxygen reduction reaction (ORR) may be responsible for such deviations[40,49]. Due to the formation of co-adsorbed oxygen spectator species under reaction conditions, the oxophilicity (i.e., the oxidation state) of the metal center is altered which is not accounted for in scaling relations derived on clean surfaces. We recently showed how both co-adsorbed species may actively participate in the surface chemistry[40], demonstrating the "non-innocence" of the spectator species. By considering a more comprehensive reaction network where both co-adsorbed species may actively participate in the reaction, we identified novel reaction pathways responsible for increased ORR activity. While *in situ* co-adsorption behavior has been studied in the context of the ORR, more research is needed to explore this phenomenon for other surface chemistries on graphene based as well as other 2D materials.

To this end, we model the oxygen evolution reaction (OER) on 12 different transition metals (Au, Co, Cr, Cu, Fe, Ir, Mn, Ni, Pd, Pt, Rh, and Ru) known to form stable single-atom catalysts in fourfold N-substituted double carbon vacancies in graphene. Based on our previous potential-dependent microkinetic model for the ORR[40], we develop a pH-dependent microkinetic model using energetics derived from density functional theory (DFT) to obtain relevant catalytic properties for the OER such as onset potential, potential-dependent coverages, and active reaction pathways.

Through this analysis, we find that all transition metals considered, proceed through co-adsorbed mechanisms near the onset potential. These novel pathways facilitate deprotonation of surface intermediates, thereby lowering predicted onset potentials when compared to the conventionally investigated "single-adsorbate" mechanisms. Similar to our previous findings for the ORR[40], we find that thermodynamic predictions alone systematically underestimate onset potentials by assuming that all elementary steps must be exergonic and ignoring temperature effects. Additionally, we show that a microkinetic analysis, considering the kinetics, reaction conditions, and coverage effects, is essential for identifying dominant reaction pathways that are not intuitive based on thermochemistry alone. Additionally, we find that the dominant reaction pathways depend on both the metal center as well as the applied potential, and as such, scaling relations derived from non-decorated metal centers may not be appropriate when evaluating graphene-based systems and other 2D materials. These findings suggest that the presence of co-adsorbed species can be significant when evaluating the catalytic performance of graphene-based single-atom catalysts, and that the *in-situ* formation of co-adsorbed reaction intermediates might be common to other 2D materials.

Computational Details

DFT Calculations

Spin-polarized plane-wave DFT-based calculations were performed using the Vienna *Ab initio* Simulation Package (VASP)[52,53]. Core-electron interactions were approximated using projector-augmented wave (PAW)[54] potentials. Exchange correlation energies were obtained from the generalized gradient approximation (GGA) using the functional by Perdew, Burke, and Ernzerhof (PBE)[55] with an energy cutoff of 600 eV. Structures were optimized until total energies were converged to 10^{-6} eV and forces acting on relaxed atoms were below 0.02 eV/Å.

Gibbs free energies were computed by adding electronic energies (E_{elec}), zero-point vibrational energies (E_{ZPVE}), and entropy (S) contributions at 298 K and 0.1 MPa (Eq. 1). E_{ZPVE} and S were calculated from harmonic frequencies that were obtained by diagonalizing the Hessian matrix with respect to the three Cartesian degrees of freedom of each atom of the adsorbed intermediates. The computational hydrogen electrode formalism was used to compute potential-dependent reaction energies[56]. We note that all calculations were performed with a fixed number of electrons during optimization and not under constant potential.

$$G = E_{elec} + E_{ZPVE} - TS \tag{1}$$

Graphene films were modeled using a *p*(6x5) supercell with a C-C distance of 1.42 Å, which is in good agreement with the experimental value of 1.42 Å [57]. For all calculations, a 15 Å vacuum layer was used to separate periodic images. Single atoms were modeled as isolated transition metal atoms supported in a double-carbon vacancy coordinated by four nitrogen atoms ($Me_1/GN_4\text{-}V_{2C}$). We previously discussed the stability of single atom transition metals in these $GN_4\text{-}V_{2C}$ sites and showed that metal cluster growth is unfavorable at low coverages[47,58]. The Brillouin zone was sampled using a $4 \times 4 \times 1$ Monkhorst-Pack *k* point mesh[59]. Gas-phase energetics were computed using only the Γ point for molecules inside a $12 \times 12 \times 12$ Å^3 box.

Microkinetic Modeling

Previous computational work performed on graphene-based single atom catalysts suggests that the oxygen reduction reaction (ORR) proceeds through the associative mechanism[46], and we therefore assume that the OER proceeds through the reverse mechanism on these single-site catalysts. Oxygen evolution was modeled using the adsorption and sequential deprotonation of water to generate O_2. Similar to the pH-dependent model developed by Rajan *et al.*, we introduce pH effects by explicitly considering both acidic and alkaline reaction steps[60]. To do this, we allow each intermediate to react with either protons or hydroxide ions from solution. Written in the acidic notation, the following steps were considered: (i) H_2O adsorption, (ii) OH* formation, (iii) O* formation, (iv) H_2O adsorption along with OOH* formation, (v) O_2 formation, and (vi) O_2 desorption where * denotes an active site.

(i) $H_2O(aq) + * \rightarrow H_2O^*$

(ii) $H_2O^* \rightarrow OH^* + (H^+ + e^-)$

$$\text{(iii)} \quad OH^* \rightarrow O^* + (H^+ + e^-)$$
$$\text{(iv)} \quad O^* + H_2O(aq) \rightarrow OOH^* + (H^+ + e^-)$$
$$\text{(v)} \quad OOH^* \rightarrow O_2^* + (H^+ + e^-)$$
$$\text{(vi)} \quad O_2^* \rightarrow O_2 + *$$

Similarly, for all electrochemical steps (ii-v), intermediates were also allowed to react with hydroxide ions (vii-x).

$$\text{(vii)} \quad H_2O^* + (OH^- - e^-) \rightarrow OH^* + H_2O(aq)$$
$$\text{(viii)} \quad OH^* + (OH^- - e^-) \rightarrow O^* + H_2O(aq)$$
$$\text{(ix)} \quad O^* + (OH^- - e^-) \rightarrow OOH^*$$
$$\text{(x)} \quad OOH^* + (OH^- - e^-) \rightarrow O_2^* + H_2O(aq)$$

Through this formalism, either H^+ or OH^- may act as charge carriers based on the activity of each species in solution. Unless otherwise noted, all onset potentials are reported at a pH of 0. We note that although electrochemical barriers are expected to have a pH-dependence, due to the lack of experimental data for these quantities, we assume barriers to be invariant with respect to pH. Once the pH-dependence of electrochemical barriers is established in the literature, future studies can readily utilize our microkinetic modeling approach to investigate the impact of pH-dependent energetics on the on the reaction mechanism and overpotential. Overpotential sensitivity to changes in electrochemical barriers are reported in Table S3 for select metals.

Similar to our methodology developed for the ORR[40], we consider a fully interlinked mechanism of co-adsorbed intermediates. At every stage of the reaction network, we allow for the adsorption/desorption and electrochemical oxidation of reaction intermediates on both sides of the plane defined by the metal-doped graphene. This fully interlinked mechanism allows us to explore a dynamic active site where co-adsorbed intermediates may act as both spectators and reactive species at every stage of the reaction. In our approach, both sides of the graphene film are considered symmetrically equivalent such that structures A*/B* and B*/A* are identical. Expanding upon our previous model[40], we consider co-adsorbed pathways for all metal centers and explicitly account for the adsorption of H_2O. In addition, structures in which multiple intermediates adsorb on one side of the graphene plane through uniplanar co-adsorption were also considered. Structures with multiple adsorbates on the same side of the graphene film are denoted as A*-B*. Representative structures for each adsorption type are shown in Figure 1. The combination of all possible double-sided (DS) and same-sided (SS) adsorption pathways is referred to as $\mathbf{M_{full}}$ (Figure 2). Additional information regarding elementary steps and rate constants is included in the Supporting Information along with full details of our microkinetic modeling formulation.

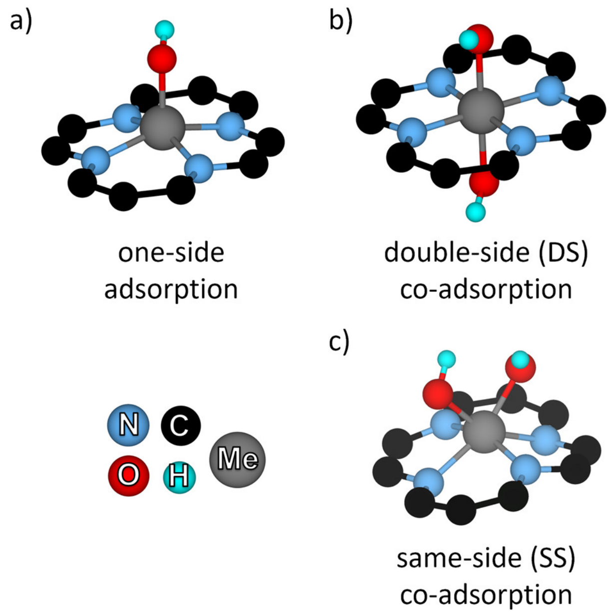

Figure 1. Representative structures demonstrating a) one-side adsorption (Me-OH), b) double-side (DS) co-adsorption (Me-OH/OH), and c) same-side (SS) co-adsorption (Me-OH-OH). Energetics for one-side adsorption and double-side co-adsorption are provided in Table S1 whereas same-side energetics are included in Table S2.

For the oxygen evolution reaction, onset potentials are often reported in reference to a current density of 10 mA cm^{-2}. Therefore, unless otherwise noted, all onset potentials presented in this work are defined in reference to a current density of 10 mA cm^{-2}. This is approximately the anodic current density expected in a 10% efficient water-spitting device under 1 sun illumination (defined as solar irradiance at the AM (air mass) 1.5 G (global) condition amounting to 1 kW/m^2)[61–63]. Through our microkinetic modeling formulation, we compute the number of electrons generated at a given potential. Using a site density of 10^{16} metal atoms cm^{-2} in line with recent experimental reports for graphene-based single-atom catalysts[26,31], we then compute current densities as a function of the applied potential. For all electrochemical reactions, a deprotonation barrier of 0.26 eV was used[40,49,64]. We note that while the predicted onset potentials have some variance with respect to these parameters, overall trends and dominant reaction mechanisms remain unchanged. As the choice of electrochemical barrier and site density affects all metals, our results are qualitatively consistent. Additional discussion regarding the onset potential sensitivity is provided in Supporting Information.

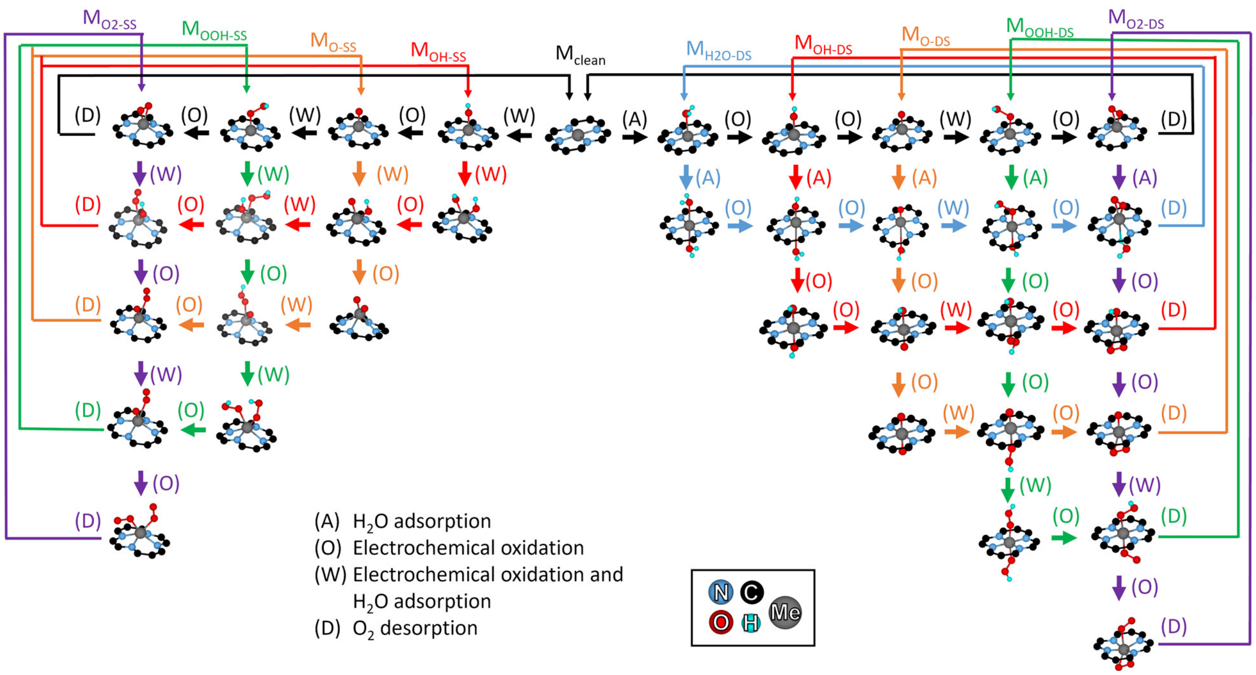

Figure 2. Full OER mechanism (**M_{Full}**) considered in this work. Labeled pathways correspond to mechanisms where each intermediate acts as a spectator species. For example, through the **$M_{OH\text{-}DS}$** pathway, a single OH species would act as a spectator on the metal center while the reaction proceeds on the other side of the graphene plane. At any stage in the reaction network, intermediate species on either side of the metal center are allowed to react. Pathways in which adsorbates co-adsorb on the same side of the graphene plane are denoted with SS (e.g., **$M_{OH\text{-}SS}$**). Pathways in which adsorbates co-adsorb through double-sided structures in which adsorbates are present on both sides of the graphene plane are denoted with DS (e.g., **$M_{OH\text{-}DS}$**)

Results and Discussion

Among transition metals, nitrogen-coordinated Co single atoms supported in carbonaceous materials have shown high activity in alkaline conditions for the oxygen evolution reaction with overpotentials as low as 0.30 V (0.1 M KOH)[65]. In agreement, computational studies also predict Co to lie near the OER activity peak[43,45,47,66]. Accordingly, we begin our analysis with Co by investigating the traditional single-adsorbate mechanism (**M_{clean}**) and our full co-adsorbed reaction network (**M_{full}**; Figure 2) using both a purely thermodynamic model as well as our pH-dependent microkinetic model.

In the absence of co-adsorbed species (**M_{clean}**), we predict an overpotential of 0.55 V for Co through a purely thermodynamic analysis with the deprotonation of hydroxyl ($OH^* \rightarrow O^* + H^+ + e^-$; $\Delta G = 1.62$ eV at U = 0 V_{RHE}) being potential-limiting (Figure 3). Using the same DFT-derived energetics, our microkinetic model predicts an enhancement in OER activity with an estimated overpotential of only 0.25 V (Figure 4). Within the framework of purely thermodynamic models, it is assumed that all elementary steps must be exergonic. However, at 298 K, temperatures effects provide enough energy for mildly endergonic steps to be overcome. As noted in our previous work on the ORR[40], the inclusion of these thermal energy contributions leads to an enhancement in activity that is not captured in purely thermodynamic predictions.

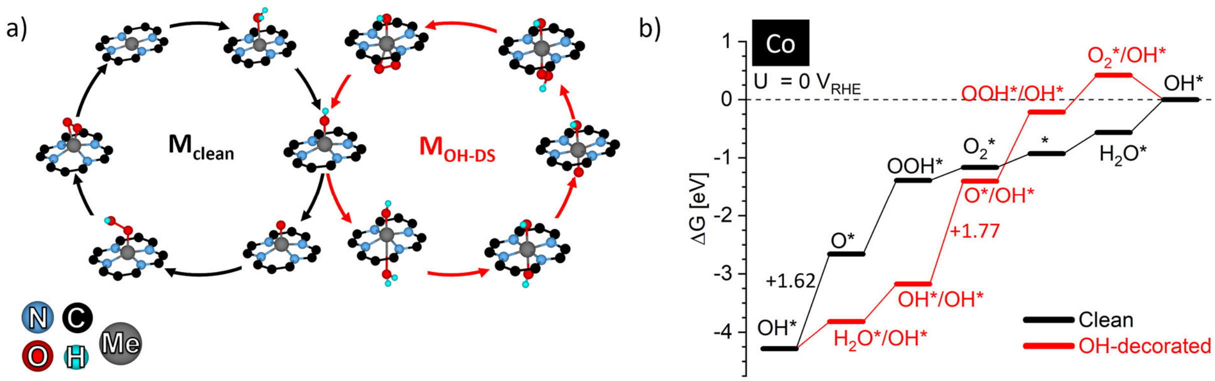

Figure 3. (a) Active OER reaction pathways for $Co_1/GN_4\text{-}V_{2C}$ comprised of clean (**M_{clean}**) and double-sided OH-decorated (**$M_{OH\text{-}DS}$**) reaction mechanisms and (b) corresponding free energy diagrams at U = 0 V_{RHE} and a pH of 0. Reaction free energies for the most endergonic elementary step are included for both mechanisms.

Upon implementation of the full reaction network (**M_{full}**) in the microkinetic model, the OER is predicted to proceed exclusively through a double-sided OH-decorated mechanism (**$M_{OH\text{-}DS}$**) on Co near the onset potential in which a spectator OH* species interacts with the metal center while the reaction proceeds on the metal center on the other side of the graphene plane (Figure 3a). After allowing for co-adsorbed states, the predicted overpotential of Co increases slightly to 0.32 V (Figure 4). Despite **M_{clean}** predicted to be more active than **$M_{OH\text{-}DS}$**, the deprotonation of a second water molecule to form a OH*/OH* state is lower in energy than deprotonating OH* to form atomic oxygen (Figure 3b), and the reaction proceeds through this alternate pathway. Based solely on a thermodynamic analysis, it is not immediately intuitive that the reaction would proceed through the less active OH-decorated mechanism. However, near the onset potential, this biplanar OH*/OH* state is more stable than a single OH* thereby leading to flux exclusively through this **$M_{OH\text{-}DS}$** mechanism. These results illustrate how surface coverage under reaction conditions is dictated by both thermodynamics and reaction kinetics. Only by directly considering all possible reaction pathways and kinetic contributions through microkinetic modeling can these trends be observed. Similar behavior was also observed for Ir and Rh where although the clean mechanism is predicted to be more active (i.e., lower overpotential), the dominant reaction flux proceeds through alternate mechanisms with lower activity (i.e., higher overpotentials) due the increased stability of co-adsorbed states (Table 1).

Table 1. Overpotentials (η) for single atom transition metals supported in fourfold N-substituted double carbon vacancies in graphene through clean and co-adsorption (**M$_{full}$**) mechanisms based on thermodynamic and microkinetic modeling (MKM) predictions. Thermodynamic predictions (η_{thermo}) for **M$_{full}$** are based on dominant reactions pathways observed from MKM (Table 2). All MKM predicted overpotentials (η_{MKM}) are defined relative to a current density of 10 mA cm^{-2} and are reported in V. When only considering the clean mechanism, O_2-desorption is rate-limiting on Cr, Ru, and Mn atoms. As this step is independent of the applied potential, these systems are unable to achieve relevant current densities, and thus, the clean mechanism is predicted to be inactive for the case of Cr, Ru, and Mn. Experimental overpotentials (η_{expt}) to achieve 10 mA cm^{-2} and the corresponding experimental conditions and active site structure are included for reference.

Metal	η_{thermo} [Clean]	η_{thermo} [Full]	η_{MKM} [Clean]	η_{MKM} [Full]	η_{expt}	pH	Expt. Conditions	Expt. Active Site
Au	1.09	0.96	0.80	0.51	-	0	-	-
	1.09	0.96	0.77	0.47	-	14	-	-
Co	0.55	0.70	0.25	0.32	-	0		
	0.55	0.70	0.23	0.31	0.30[65], 0.31[19]	13	0.1 M KOH	CoN_4
	0.55	0.70	0.21	0.29	0.40[26]	14	1.0 M KOH	CoN_4
Cr	2.00	1.41	Inactive	0.81	-	0	-	-
	2.00	1.41	Inactive	0.78	-	14	-	-
Cu	0.75	0.78	0.52	0.48	-	0	-	-
	0.75	0.78	0.48	0.44	-	14	-	-
Fe	1.05	0.71	0.46	0.24	0.29[28]	0	0.5 M H_2SO_4	FeN_4
	1.05	0.71	0.43	0.20	0.34[27], 0.43[3], 0.49[26]	14	1.0 M KOH	FeN_4
Ir	0.67	0.51	0.11	0.19	0.27[32], 0.32[67]	0	0.5 M H_2SO_4	IrN_4
	0.67	0.51	0.08	0.16	-	14	-	-
Mn	1.39	0.95	Inactive	0.36	-	0	-	-
	1.39	0.95	Inactive	0.32	0.34[29]	14	1.0 M KOH	MnN_4
Ni	0.98	0.72	0.70	0.63	0.40[34]	0	0.5 M H_2SO_4	NiN_4
	0.98	0.72	0.66	0.59	0.30[34], 0.33[26], 0.36[27]	14	1.0 M KOH	NiN_4
Pd	0.95	1.06	0.90	0.71	-	0	-	-
	0.95	1.06	0.87	0.67	-	14	-	-
Pt	0.97	0.94	0.91	0.64	-	0	-	-
	0.97	0.94	0.87	0.60	-	14	-	-
Rh	0.50	0.63	0.23	0.30	-	0	-	-
	0.50	0.63	0.20	0.27	-	14	-	-
Ru	1.09	0.95	Inactive	0.35	0.27[31]	0	0.5 M H_2SO_4	RuN_4
	1.09	0.95	Inactive	0.32	-	14	-	-

As most OER experiments are performed under alkaline conditions to enhance catalyst stability, we also performed our microkinetic analysis at various pH conditions. Our results show that increasing the pH alters the dominant charge carrier from H^+ to OH^-, but it does not alter dominant reaction mechanisms or the predicted coverages. We find that predicted onset potentials for all metals decrease as the pH is increased. As H^+ is a product of acidic reaction steps (ii-v), increasing the concentration of H^+ (lowering pH) corresponds to increasing the reverse reaction rates. In contrast, OH^- acts as a reactant for alkaline steps (vii-x) and increasing the concentration of OH^- (increasing pH) corresponds to increasing the forward rate of alkaline steps. Through our formulation, this leads to an increase in activity for all alkaline steps over their acidic counterparts. As changes in pH were not found to alter dominant reaction mechanisms, rate-determining steps, or surface coverages, we do not include any additional discussion. Onset potential for acidic and alkaline conditions for all metals are shown in Figure 6 and numerical values are reported in Table 1.

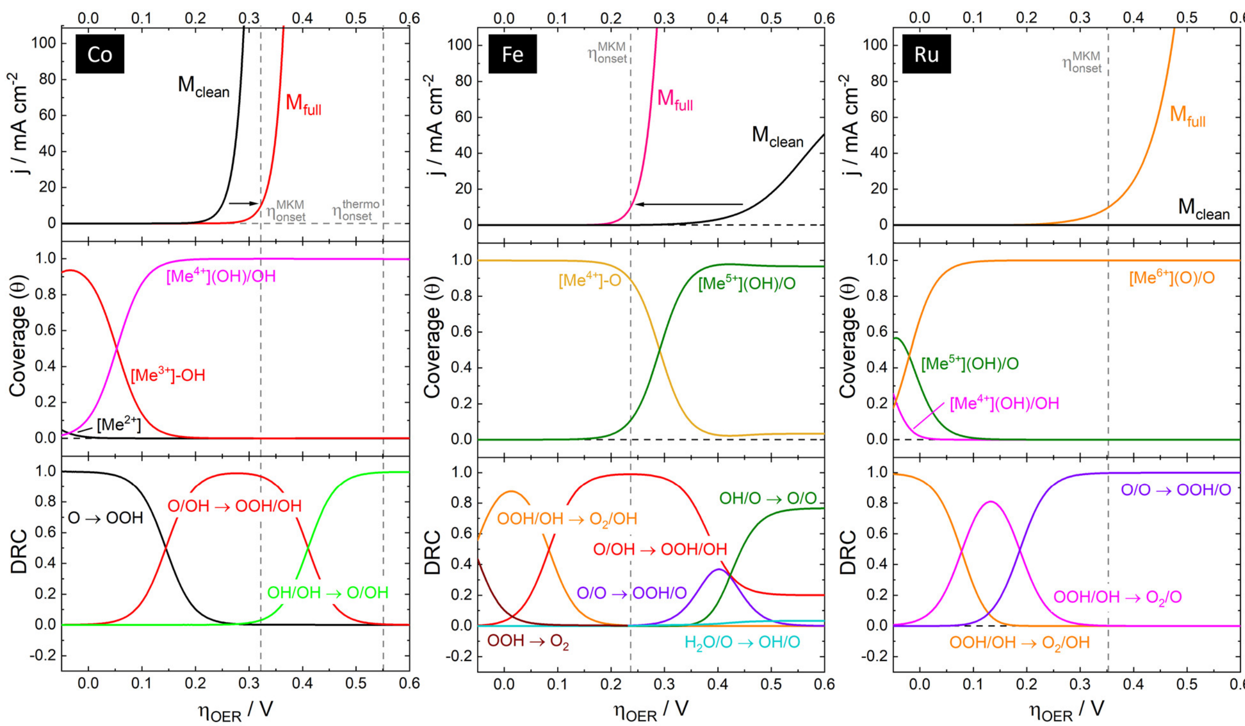

Figure 4. Current density, site coverage, and degree of rate control (DRC) all calculated from microkinetic analysis of OER on Co, Fe, and Ru single-atom catalysts at a pH of 0 and temperature of 298 K. Black arrows indicate the shift in current response from the clean mechanism (**M_{clean}**) to the full mechanism (**M_{full}**). Dashed vertical lines denote the predicted onset potential from microkinetic modeling (η_{onset}^{MKM}) corresponding to a current density of 10 mA cm^{-2} and traditional thermodynamic predictions for **M_{clean}** (η_{onset}^{thermo}). Coverage and DRC profiles are derived using **M_{full}**. For Fe and Ru, purely thermodynamic predictions estimate overpotentials greater than 1.0 V (Table 1). Coverage profiles for only the relevant states are shown in the respective panel for each metal. The current density of Ru through the clean mechanism is predicted to be inactive as it is limited by the O_2 desorption. Cation charges (Me^{N+}) represent formal oxidation states; Bader charges are reported in Table S5.

While experimental measurements and computational predictions agree on the high activity of Co, they deviate more substantially when considering other transition metals. For example, a thermodynamic analysis from our DFT energetics predicts Fe to require an overpotential of 1.05 V before the OER becomes active. However, high OER activity has been observed for Fe single atoms in both acidic and alkaline media with overpotentials as low as 0.29 V (0.5 M H_2SO_4)[28] and 0.34-0.49 V (1.0 M KOH)[3,26,27], respectively. These experiments indicate that the activity of Fe is far closer to Co than purely thermodynamic predictions suggest. Upon implementation of our full microkinetic model, we find that Fe, like Co, proceeds exclusively through a decorated mechanism near the onset potential. However, unlike on Co, OH* does not act solely as a spectator species on Fe. Under the clean mechanism ($\mathbf{M}_{\mathbf{clean}}$), Fe is limited by O* reacting with a second water molecule to generate OOH* (O* + H_2O(aq) → OOH* + H^+ + e^-, ΔG = 2.12 eV at U = 0 V_{RHE}). When considering the full mechanism ($\mathbf{M}_{\mathbf{full}}$; Figure 2), instead of O* reacting, a second water molecule can adsorb on the clean facet of the metal center and deprotonate to generate a OH* species (O* + H_2O(aq) → O*/H_2O* → O*/OH* + H^+ + e^-; ΔG = 1.34 eV at U = 0 V_{RHE}). Upon the co-adsorption of OH*, the deprotonation of a second water molecule is facilitated (O*/OH* + H_2O(aq) → OOH*/OH* + H^+ + e^-; ΔG = 1.79 eV at U = 0 V_{RHE}), and a lower onset potential is observed. Near the onset potential, the Fe single atom is predicted to be decorated by O* and proceed primarily through this biplanar OH-assisted mechanism ($\mathbf{M}_{\mathbf{OH\text{-}astd\text{-}1}}$, Table 2). Moving towards higher potentials, the emergence of a co-adsorbed hydroxyl species (O*/OH*) is predicted to dominate the surface coverage leading to a higher oxidation state (Figure 4) with additional contributions from the O-decorated ($\mathbf{M}_{\mathbf{O}}$) mechanism contributing to the overall reaction flux.

Table 2. Active OER pathways identified from microkinetic modeling for each single atom transition metal near the onset potential (10 mA cm^{-2}). For clarity, proton-electron pairs and adsorbed/desorbed species are excluded from the reaction pathway notation. Species separated by a slash (e.g., OH*/OH*) denote co-adsorbed species one on each side of the graphene plane whereas species separated by a dash (e.g., OH*-OH*) indicate same-side (SS) co-adsorption.

Metal(s)	Reaction Pathway	Notation
Au, Cu	OH* → OH*-OH* → O*-OH* → OOH*-OH* → O_2*-OH* → OH*	$\mathbf{M}_{\mathbf{OH\text{-}SS}}$
Co, Ni, Pd, Pt	OH* → H_2O*/OH* → OH*/OH* → O*/OH* → OOH*/OH* → O_2*/OH* → OH*	$\mathbf{M}_{\mathbf{OH\text{-}DS}}$
Fe	OH* → O* → H_2O*/O* → OH*/O* → OH*/OOH* → OH*/O_2* → OH*	$\mathbf{M}_{\mathbf{OH\text{-}astd\text{-}1}}$
Ir	O* → H_2O*/O* → OH*/O* → OH*/OOH* → OH*/O_2* → O*/O_2* → O*	$\mathbf{M}_{\mathbf{OH\text{-}astd\text{-}2}}$
Mn, Rh, Ru	O* → H_2O*/O* → OH*/O* → O*/O* → OOH*/O* → O_2*/O* → O*	$\mathbf{M}_{\mathbf{O\text{-}DS}}$
Cr	O*-O* → O*-O*-OH* → O*-O*-O* → O*-O*-OOH* → O*-O*-O_2* → O*-O*	$\mathbf{M}_{\mathbf{2O\text{-}SS}}$

Moving towards more oxophilic metals, a purely thermodynamic analysis using our DFT-derived energetics predicts Ru and Mn to be poor OER electrocatalysts with overpotentials in excess of 1 V. Based on these predictions, Ru and Mn are estimated to possess onset potentials 0.54 V and 0.83 V higher than Co, respectively (Table 1). However, recent experimental evidence for Ru and Mn single-atom catalysts supported on N-doped graphene has shown onset potentials lower than or comparable to Co catalysts[29,31]. Our microkinetic model reveals that when considering only the clean mechanism, both Ru and Mn are inactive towards the OER. As the overall reaction rate is limited by the desorption of molecular oxygen, a step independent of the applied potential, appreciable current densities cannot be achieved through this mechanism alone due to poisoning of the active site. However, by enabling the formation of co-adsorbed states, significant enhancements in activity are observed. Starting with Ru, the inclusion of the full mechanism leads to an overpotential of only 0.35 V to achieve a current density of 10 mA cm^{-2} which is in good agreement with the experimental observation of 0.27 V[31] (0.5 M H_2SO_4). At relevant potentials, Ru is predicted to proceed exclusively through a biplanar O-decorated mechanism ($\mathbf{M_O}$) with adsorbed O* reacting with a second water molecule to form OOH* ($O^*/O^* + H_2O(aq) \rightarrow OOH^*/O^* + H^+ + e^-$; $\Delta G = 2.02$ eV at $U = 0\ V_{RHE}$) possessing the highest degree of rate control (Figure 4; purple line). These results are consistent with recent experimental work where Cao *et al*. observed the emergence of a reversible FTIR peak near 1.5 V_{RHE} attributed to the formation of an adsorbed oxo species under reaction conditions on Ru-N-C catalysts[31]. A similar phenomenon was also observed by Guan *et al*. on Mn single atoms where high OER activity was attributed to the formation of an adsorbed oxo species on the metal center leading to an increase in the Mn oxidation state[29]. In agreement, our microkinetic model predicts that Mn single atoms also proceed exclusively through $\mathbf{M_O}$ with a low overpotential of only 0.36 V. For both Ru and Mn atoms, the presence of a co-adsorbed O* species elevates the oxidation state of the metal center leading to significant enhancement in activity compared to the undecorated site.

Interestingly, our microkinetic model predicts that all metals studied proceed through decorated mechanisms (Table 2). These findings are in contrast to our previous work on the ORR where decorated mechanisms were only active on the most oxophilic metals (i.e., Fe, Mn, Ru, and Cr)[40]. Even on weakly oxophilic metals (i.e., Co, Ni, Pd, Pt), we find that the dominant reaction flux at the onset potential proceeds exclusively through a biplanar OH-decorated mechanism ($\mathbf{M_{OH}}$). On these systems, OH* serves primarily as a spectator that mediates the oxidation state of the metal center and modifies the binding of reactive intermediates. Both Au and Cu single atoms were also found to proceed through OH-decorated mechanisms. However, for these metals the co-adsorption of multiple intermediates on one side of the graphene plane was found to be energetically preferred. Upon the adsorption of a second intermediate, the metal center was found to tilt out of the graphene plane forming an N-undercoordinated structure where the metal center becomes coordinated by two-nitrogen atoms and two OER intermediates. Based on the lack of experimental evidence for the OER activity of CuN_4 and AuN_4 catalysts, materials in which the formation of N-undercoordinated metal centers is energetically favorable may be unstable at potentials relevant for the OER. Reaction mechanisms and corresponding free energy diagrams for Cu are shown in Figure 5. In fact, for all late transition metals (Au, Cu, Ni, Pd, Pt), uniplanar co-adsorption (e.g., OH*-OH*) was found to lead to the spontaneous formation of N-undercoordinated metal centers. Such behavior has been suggested previously based on experimental evidence for Cu and Ni single atoms catalysts under ORR conditions[68,69]. While we only predict these N-undercoordinated structures to be OER active for Cu and Au, additional considerations such as solvent effects may stabilize uniplanar co-adsorption on Ni, Pd, and Pt and alter dominant reaction pathways.

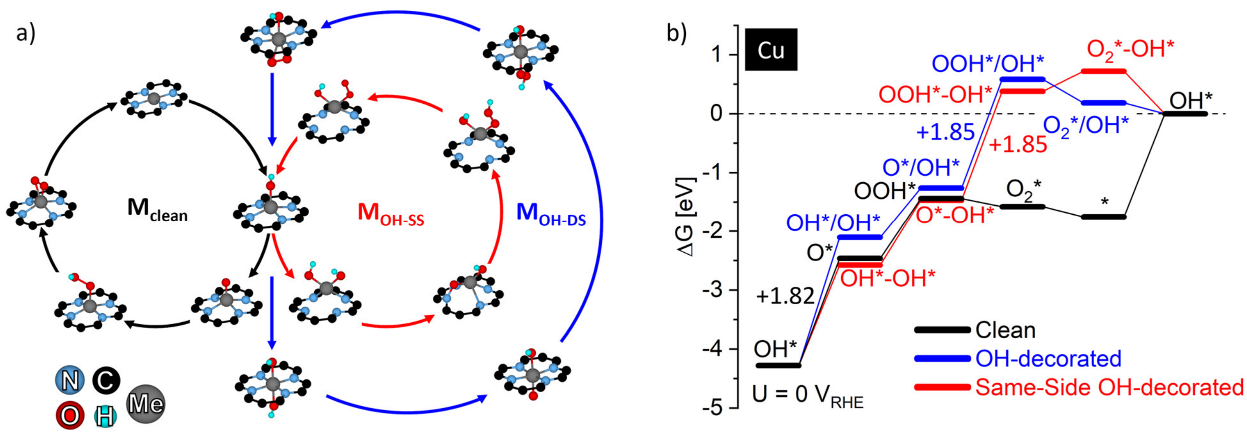

Figure 5. (a) OER reaction mechanisms and (b) corresponding free energy diagram at U = 0 V_{RHE} and a pH of 0 for clean (**M_{clean}**, black), double-sided OH-decorated (**M_{OH}**, blue), and same-sided OH-decorated (**M_{OH-SS}**, red) mechanisms over $Cu_1/GN_4\text{-}V_{2C}$. Reaction free energies for the most endergonic elementary step are included for each mechanism.

When comparing reaction mechanisms between the metals investigated, we find that the oxophilicity of the metal center dictates the dominant reaction mechanism. Weakly oxophilic metals (Au, Co, Cu, Ni, Pd, Pt) were found to proceed through OH-decorated mechanisms. For Group 9 and 10 elements (Co, Ni, Pd Pt), a biplanar OH-decorated mechanism (**M_{OH}**) was found to be dominant where OH* acts as spectator species, and the OER proceeds on the other side of the graphene plane. In contrast, both Au and Cu were found to proceed through a uniplanar OH-decorated mechanism (**M_{OH-SS}**) where again, OH* acts as a spectator, but the reaction now proceeds on the same side of the graphene plane. Transitioning to more oxophilic metals, we find that Fe and Ir proceed through mechanisms where both OH* and O* act as spectators during various stages of the reaction. Next, Mn, Ru, and Rh proceed exclusively through an O-decorated mechanism (**M_O**) where O* alone acts as the spectator. When taken to the extreme, we find that for the most oxophilic metal considered (Cr), the adsorption of a third intermediate is energetically preferred leading to a dominant reaction mechanism where multiple O* species act as spectators. Structures for this **M_{2O-SS}** mechanism along with additional discussion are included in Supporting Information. This general behavior is consistent with expected oxidation trends for each metal center. To illustrate this, we consider the binding energy of atomic oxygen (BE_O) referenced to gas phase O_2 on the undecorated metal centers as a proxy for how easily each metal center can enter a higher oxidation state. When comparing between metal centers, the following trend between dominant reaction pathways and BE_O on the undecorated metal centers is observed: Cu (**$M_{OH\text{-}SS}$**; BE_O = 1.09 eV) > Co (**$M_{OH\text{-}DS}$**; BE_O = 0.16 eV) > Fe (**$M_{OH\text{-}astd\text{-}1}$**; BE_O = -1.21eV) > Ru (**$M_{O\text{-}DS}$**; BE_O = -1.75 eV) > Cr (**$M_{2O\text{-}SS}$**; BE_O = -2.66 eV). For metals such as Cu, where elevating the oxidation state is the most difficult (i.e., high BE_O), the formation of a N-undercoordinated active site which minimizes the increase in oxidation state of the metal center is observed. For metals where oxidation is more attenable, the formation of biplanar co-adsorbed states with either OH* (for intermediate BE_O values) or O* (for more negative BE_O values) is preferred. Dominant reaction mechanisms for each system are shown in Table 2. A Bader charge analysis for select metals is included in Supporting Information. Our results indicate that under reaction conditions, the active site of these graphene-based single atom catalysts is dynamic, depending on both the applied potential and the identity of the metal center. Due to the dynamic nature of the active site, predicting active reaction pathways and subsequent onset potentials from purely thermodynamic analyses remains challenging. Only through microkinetic modeling, where all reaction pathways are permitted, can insights about active reaction pathways be derived.

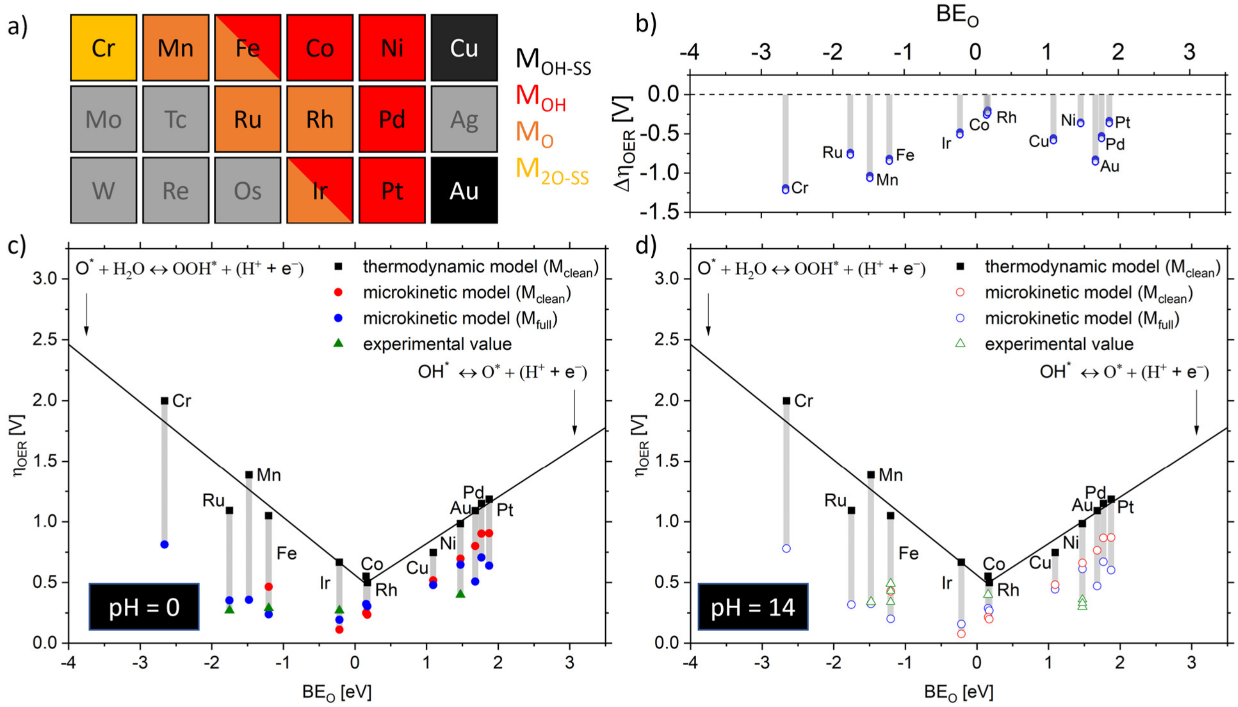

Figure 6. (a) Dominant reaction pathways near the onset potential for each metal. Boxes composed of multiple colors indicate crossover of distinct decorated pathways. Grey boxes denote systems that were not considered as they are predicted to be unstable. (b) Differences in the overpotentials predicted from thermodynamic and microkinetic analysis ($\Delta\eta_{OER} = \eta_{onset}^{MKM} - \eta_{onset}^{thermo}$): negative values correspond to a decrease in overpotential due to the microkinetic analysis. Onset potentials for single atom transition metals supported in N-doped graphene (Me_1/GN_4-V_{2C}) at a pH of (c) 0 and (d) 14. OER onset potentials from thermodynamic analysis (black squares; linear fit shown as black line) and from microkinetic analysis (colored circles). Blue and red circles correspond to onset potentials calculated at 10 mA cm^{-2} for the **M$_{clean}$** and **M$_{full}$**, respectively. Red points for Cr, Mn, and Ru are excluded as these metals are predicted to be inactive through the clean mechanism. Green triangles correspond to experimentally reported onset potentials. Numeric values are provided in Table 1.

OER overpotentials for all metal atoms considered in this work are shown in Figure 6 as a function of their electronic binding energy of atomic O* (BE_O) relative to gas-phase O_2. The inclusion of co-adsorbed species leads to significant broadening of the activation peak derived from a purely thermodynamic analysis which is in better agreement with experimental observations (Figure 6). With the exception of Ni, we find that under acidic conditions our microkinetic model accurately predicts onset potentials within 0.1 V of experimental values (Figure 6c) which is similar to the offset by which PBE underestimates the equilibrium potential of the OER ($\Delta U_{eqm} = 0.16$ V). Under alkaline conditions, our predicted onset potentials deviate by as much as 0.3 V (Figure 6d). However, we note that no pH-dependence on electrochemical barriers or solvent reorganization energies was included in our model due to lack of experimental data for these quantities. Moving towards more alkaline conditions, we would expect electrochemical barriers to increase and solvent reorganization to become more difficult both leading to a suppression of OER activity[60,70]. For select metals, the activation barrier of electrochemical steps was deliberately varied, and the resulting effects are included in Table S3. Ni is the one exception to this trend in which experiments suggest Ni is more active towards the OER under alkaline conditions. However, previous works have indicated that differences in N-coordination of the Ni single atom influenced by synthesis conditions[47] and other support effects[26,68] may be responsible for this discrepancy.

Conclusions

In this work, we used first-principles informed microkinetic models to study the oxygen evolution reaction on graphene-based single-atom (Me_1/GN_4-V_{2C}) catalysts. Similar to recent observations for the ORR[40,49], we find that thermodynamic models systematically underestimate onset potentials. We determined that reaction pathways are influenced by features such as metal site coverage which are not considered in purely thermodynamic models, leading to dominant pathways that are not always intuitive based on thermochemistry alone. Pathways involving species co-adsorbed on the metal center are common to graphene-based single atom catalysts, and the reaction mechanisms depend on both the identity of the metal and the applied potential. These findings show the limitations of thermodynamic models and illustrate the commonality of co-adsorbed pathways on these graphene-based materials. Accordingly, co-adsorbed species should be considered when evaluating the catalytic performance of graphene-based single-atom catalysts. Based on the commonality of co-adsorbed states on graphene-based catalysts, we suggest that such states may also be relevant for other 2D materials and should be considered explicitly when evaluating their catalytic properties.

Supporting Information

This information is available free of charge on the ACS Publications website.

Gibbs free reaction energies; microkinetic modeling; sensitivity analysis; electronic properties of metal centers for selected systems. (PDF)

Optimized structures from DFT. (ZIP)

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